## TauTimp — Temporal Wave Theory: Solar Temporal Flux and Temporal Dilation

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### 1. Variable-Coefficient Wave Equation and the H-Field 1.1 Master Wave Equation

The TauTimp framework postulates a variable-coefficient spherical wave equation for the temporal field $H(r,\theta,t)$:

$$\frac{R}{r}\,\Delta H(r,\theta,t) = \frac{1}{c^2}\frac{\partial^2 H(r,\theta,t)}{\partial t^2}$$

where $R = 4.2355 \times 10^{22}$ m is the TauTimp cosmological radius, $c = 2.998 \times 10^8$ m/s, and $\Delta$ is the Laplacian in spherical coordinates.

### 1.2 Spherical Symmetry Reduction

Under spherical symmetry, $H$ is independent of azimuthal and polar angles, so $H = H(r,t)$. The Laplacian reduces to:

$$\Delta H = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial H}{\partial r}\right) = \frac{\partial^2 H}{\partial r^2} + \frac{2}{r}\frac{\partial H}{\partial r}$$

Equation (1) becomes: $\frac{R}{r}\left(\frac{\partial^2 H}{\partial r^2} + \frac{2}{r}\frac{\partial H}{\partial r}\right) = \frac{1}{c^2}\frac{\partial^2 H}{\partial t^2}$

### 1.3 Separation of Variables

We write $H(r,t) = g(r)\,u(t)$. The temporal part satisfies:

$$\ddot{u} + \omega^2 u = 0 \Longrightarrow u(t) = e^{\pm i\omega t}$$

The radial equation becomes:

$$\frac{R}{r}\left(g'' + \frac{2}{r}g'\right) = -\frac{\omega^2}{c^2}g \tag{5}$$

### 1.4 Airy Reduction of the Radial Equation

Substituting $g(r) = \phi(r)/r$ and defining $\xi = \left(\frac{\omega^2 r}{Rc^2}\right)^{1/3}$, equation (5) transforms to: $\frac{d^2\phi}{d\xi^2} = \xi\,\phi$

This is **Airy's equation**. The general solution is: $\phi(\xi) = C_1\,\mathrm{Ai}(\xi) + C_2\,\mathrm{Bi}(\xi)$

Since $\mathrm{Bi}(\xi) \to \infty$ as $\xi \to \infty$ (physically inadmissible), we set $C_2 = 0$: $g(r) = \frac{C_1}{r}\,\mathrm{Ai}\left[\left(\frac{\omega^2 r}{Rc^2}\right)^{1/3}\right]$

### 1.5 Eigenvalue Quantization

The boundary condition $H(R,t) = 0$ requires $g(R) = 0$, i.e., $\mathrm{Ai}(\xi_R) = 0$. The Airy zeros $a_n$ ($n = 1,2,3,\ldots$) give:

$$\xi_R = a_n \Longrightarrow \omega_n = c\sqrt{\frac{R\,a_n^3}{R^3}} = \frac{c\,a_n^{3/2}}{R}$$

For the principal mode $l = 21$ (TauTimp quantum number), the relevant Airy zero is $a_{21} \approx -17.03$, yielding:

$$\omega_{21} = \frac{c\,|a_{21}|^{3/2}}{R} \approx \frac{2.998\times10^8 \times 70.33}{4.2355\times10^{22}} \approx 4.98\times10^{-13}\ \text{rad/s} \tag{10}$$

### 1.6 Full H-Field via Legendre Expansion

Restoring angular dependence via spherical harmonics $Y_l^m(\theta,\phi)$:

$$H(r,\theta,t) = \sum_{n,l,m}\frac{C_{nlm}}{r}\,\mathrm{Ai}\left[\left(\frac{\omega_n^2 r}{Rc^2}\right)^{1/3}\right]Y_l^m(\theta,\phi)\,e^{\pm i\omega_n t}$$

For the $m = 0$ (azimuthal symmetry) $l = 21$ dominant mode: $H(r,\theta,t) = \frac{C}{r}\,\mathrm{Ai}\left[\left(\frac{\omega_{21}^2 r}{Rc^2}\right)^{1/3}\right]P_{21}(\cos\theta)\cos(\omega_{21}t)$

where $P_{21}$ is the Legendre polynomial of degree 21.

*& the version with Airy function variant:*

### 1. The TauTimp Field Equation

The master equation in spherical coordinates is: $\frac{R}{r}\,\Delta H(r,\theta,t) = \frac{1}{c^2}\frac{\partial^2 H(r,\theta,t)}{\partial t^2}$

Under **spherical symmetry** ($H$ independent of $\theta$ and $\phi$), the Laplacian reduces to:

$$\Delta f = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial f}{\partial r}\right) = f_{rr} + \frac{2}{r}f_r$$

The scalar equation to solve is: $\boxed{\frac{R}{r}\left(f_{rr} + \frac{2}{r}f_r\right) = \frac{1}{c^2}\,f_{tt}, \qquad f = f(r,t)}$

### 2. Separation of Variables: $f(r,t) = g(r)\,u(t)$

Substituting and dividing by $g(r)\,u(t)$: $\frac{R}{r}\cdot\frac{g''+\frac{2}{r}g'}{g} = \frac{1}{c^2}\cdot\frac{u''}{u} = -\lambda^2, \qquad \lambda^2 > 0$

This yields two ordinary differential equations.

### 3. Temporal ODE — Exact Analytic Solution

$$u''(t) + \lambda^2 c^2\,u(t) = 0$$

$$\boxed{u(t) = A\cos(\lambda c\,t) + B\sin(\lambda c\,t), \qquad \omega = \lambda c}$$

### 4. Spatial ODE — Reduction to the Airy Equation

The radial equation is: $g''(r) + \frac{2}{r}g'(r) + \frac{\lambda^2 r}{R}\,g(r) = 0$

### Step 1 — Substitution $g(r) = v(r)/r$

Since $g'' + \frac{2}{r}g' = \frac{v''}{r}$, the equation simplifies exactly to: $\boxed{v''(r) + \frac{\lambda^2}{R}\, r \cdot v(r) = 0}$

### Step 2 — Change of variable to canonical Airy form

Set $\xi = \left(\frac{\lambda^2}{R}\right)^{1/3} r$, so that $\frac{d^2v}{d\xi^2} + \xi\, v = 0$.

This is the **standard Airy equation**, with exact solution:

$$v(\xi) = C_1\, \mathrm{Ai}(-\xi) + C_2\, \mathrm{Bi}(-\xi)$$

Since $\mathrm{Bi}(-\xi) \to \infty$ as $\xi \to 0^+$ (diverges at the origin), we set $C_2 = 0$ for regularity:

$$g(r) = \frac{v(r)}{r} = \frac{\mathrm{Ai}\left(-(\lambda^2/R)^{1/3} r\right)}{r}$$

### 5. Eigenvalue Quantization: Boundary Condition $g(R) = 0$

The condition $g(R) = 0$ requires $v(R) = 0$, i.e.: $\mathrm{Ai}\left(-\left(\frac{\lambda_n^2}{R}\right)^{1/3} R\right) = 0$

Denoting $\zeta_n = (\lambda_n^2/R)^{1/3} R$, the zeros $a_n$ of $\mathrm{Ai}(-\zeta) = 0$ are the (positive) Airy zeros:

| $n$ | $a_n$ |
|---|---|
| 1 | 2.3381 |
| 2 | 4.0879 |
| 3 | 5.5206 |
| 4 | 6.7867 |

**Discrete eigenvalues:** $\boxed{\lambda_n = \frac{a_n^{3/2}}{R}, \qquad n = 1,2,3,\ldots}$

**Spatial eigenfunctions:** $\boxed{g_n(r) = \frac{1}{r}\, \mathrm{Ai}\left(-a_n \frac{r}{R}\right)}$

### 6. Complete Solution $f(r,t)$

By superposition:

$$\boxed{f(r,t) = \sum_{n=1}^{\infty} \frac{\mathrm{Ai}(-a_n\, r/R)}{r}\left[A_n \cos\left(\frac{a_n^{3/2} c}{R}\, t\right) + B_n \sin\left(\frac{a_n^{3/2} c}{R}\, t\right)\right]}$$

Coefficients $A_n$, $B_n$ are determined by initial conditions via the weighted inner product of Airy functions.

### 7. Reconstruction of $H(r,\theta,t)$

The spherically symmetric sector ($\ell = 0$) corresponds to the spherical harmonic $Y_0^0 = 1/\sqrt{4\pi}$:

$$\boxed{H(r,\theta,t) = \frac{f(r,t)}{\sqrt{4\pi}} = \frac{1}{\sqrt{4\pi}} \sum_{n=1}^{\infty} \frac{\mathrm{Ai}(-a_n\, r/R)}{r}\left[A_n \cos(\omega_n t) + B_n \sin(\omega_n t)\right]}$$

where $\omega_n = a_n^{3/2}\, c/R$.

### 2. Temporal Probability Field P(r,t) 2.1 Governing PDE

The temporal probability amplitude $P(r,t)$ satisfies the spherical d'Alembert equation: $\frac{\partial^2 P(r,t)}{\partial t^2} - c^2 \frac{\partial^2 P(r,t)}{\partial r^2} = 0$

### 2.2 Boundary Conditions

Inner boundary — Schwarzschild threshold of the Sun ($3.8\, R_{s,\odot}$):

*$P(r_0, 0) = 0$, $r_0 = 11\,222.3$ m* Outer (cosmological) boundary: *$P(R, 0) = 0$, $R = 4.2355 \times 10^{22}$ m*

2.3 Polynomial Ansatz — Degree 2 in r and t

The TauTimp ansatz satisfying both boundary conditions by construction is:

$$\boxed{P(r,t) = A[(r - r_0)(r - R) + \varepsilon\, t + c^2\, t^2]}$$

**Verification in PDE:** Se calculează derivatele parțiale: $\frac{\partial^2 P}{\partial t^2} = 2Ac^2$ & $\frac{\partial^2 P}{\partial r^2} = 2A$ ; $2Ac^2 - c^2 \cdot 2A = 0$ ✓

**Verification of boundary conditions:** $P(r_0, 0) = A[(r_0 - r_0)(r_0 - R)] = 0$ ✓

$P(R, 0) = A[(R - r_0)(R - R)] = 0$ ✓

### 3. Explicit Polynomial Form and Coefficients

Expanding $(r - r_0)(r - R) = r^2 - (r_0 + R)r + r_0 R$: $P(r,t) = A\, r^2 - A(r_0 + R)\, r + A\, r_0 R + A\varepsilon\, t + Ac^2\, t^2$

| Coefficient | Analytical Expression | Value (SI) | SI Dimension |
|---|---|---|---|
| $C_{r^2} = A$ | $A$ | $9.5510 \times 10^{-18}$ | $\mathrm{m}^{-2}$ |
| $C_{r^1} = -A(r_0 + R)$ | $-A(r_0 + R)$ | $-4.0453 \times 10^{5}$ | $\mathrm{m}^{-1}$ |
| $C_{r^0} = Ar_0R$ | $A\, r_0\, R$ | $4.5398 \times 10^{9}$ | dimensionless |
| $C_{t^1} = A\varepsilon$ | $A\varepsilon = \tau_q c^2$ | $1.5863 \times 10^{-2}$ | $\mathrm{m}^2/\mathrm{s}$ |
| $C_{t^2} = Ac^2$ | $Ac^2$ | $8.5845 \times 10^{-1}$ | $\mathrm{m}^2/\mathrm{s}^2$ |

## 4. Determination of Coefficients A and ε 4.1 Time Quantum Constraint (Earth)

The initial temporal rate is linked to the Earth's time quantum $\tau_q = 1.765 \times 10^{-19}$ s: $\left.\frac{\partial P}{\partial t}\right|_{t=0} = A\varepsilon = \tau_q\, c^2$

$A\varepsilon = 1.765\times10^{-19} \times (2.998\times10^{8})^2 = 1.5863\times10^{-2}$ m²/s

### 4.2 GEO Satellite Calibration Constraint

The temporal probability flux (dimensionless) is defined as: $F(r,t) = \frac{P(r,t)}{4\pi r^2}$

The calibration condition for GEO⁺ satellites (facing away from Sun) reads:

$$\Delta F_{\text{GEO}^+} = F(r_{\text{GEO}^+},\, t_0) - F(r_\oplus,\, t_0) = +0.22 \times 10^{-6}\ \text{hr}^{-1}$$

with $r_\oplus = 1.49 \times 10^{11}$ m, $r_{\text{GEO}^+} = r_\oplus + 42\,157$ km $= 1.4942 \times 10^{11}$ m, $t_0 = 500$ s. Let $Q(r) = (r - r_0)(r - R)$:

$$\Delta F = \frac{1}{4\pi}\left\{A\left[\frac{Q(r_{\text{GEO}^+})}{r_{\text{GEO}^+}^2} - \frac{Q(r_\oplus)}{r_\oplus^2}\right] + A\varepsilon\, t_0\left[\frac{1}{r_{\text{GEO}^+}^2} - \frac{1}{r_\oplus^2}\right] + Ac^2 t_0^2\left[\frac{1}{r_{\text{GEO}^+}^2} - \frac{1}{r_\oplus^2}\right]\right\}$$

Solving the system (23) + (26): $\boxed{A = 9.5510 \times 10^{-18}\ \text{m}^{-2}}$

$$\boxed{\varepsilon = \frac{\tau_q c^2}{A} = 1.6610 \times 10^{15}\ \text{m}^2/\text{s}}$$

## 5. Complete Solution with Numerical Coefficients 5.1 SI Units

$$\boxed{P(r,t) = 9.5510 \times 10^{-18} \cdot [(r - 11\,222.3)(r - 4.2355 \times 10^{22}) + 1.6610 \times 10^{15}\, t + (2.998 \times 10^{8})^2\, t^2]}$$

### 5.2 Natural Units ($\hbar = c = 1$; using $\hbar c = 0.19733$ GeV·fm)

$$A_{\text{nat}} = A \cdot (\hbar c)^2 = 3.719 \times 10^{-49}\ \text{GeV}^2$$

ε_nat = ε/c = $5.540\times10^{6}$ m = $5.540\times10^{21}$ fm

## 6. Temporal Flux and Physical Interpretation 6.1 Temporal Probability Density

$$F(r,t) = \frac{A}{4\pi r^2}[(r - r_0)(r - R) + \varepsilon t + c^2 t^2]$$

**Physical interpretation:** - $F < 0$: time quanta absorbed (inner gravitational regime, Sun vicinity). - $F$ less negative at larger $r$: weaker temporal field, slower passage of time. - $\Delta F > 0$ for a satellite farther from Sun: positive temporal dilation (slower clock).

### 6.2 Flux at Earth ($t_0 = 500$ s — 8 min 20 s, $r_\oplus = 1.49 \times 10^{11}$ m)

$$F(r_\oplus,\, t_0) = F(1.49 \times 10^{11},\, 500) = -2.1605 \times 10^{-7}$$

### 6.3 Flux at Mars ($t_1 = 700$ s, $r_{\text{Mars}} = 2.09 \times 10^{11}$ m)

$$F(r_{\text{Mars}},\, t_1) = F(2.09 \times 10^{11},\, 700) = -1.5403 \times 10^{-7}$$

$$\Delta F_{\text{Mars}} = F(r_{\text{Mars}}, t_1) - F(r_\oplus, t_0) = +6.202 \times 10^{-8}$$

## 7. Satellite Time Variations 7.1 Orbital Parameters

| Body | $\Delta r$ from Earth | $r_{\text{Sun}}$ (m) | $t$ (s) |
|---|---|---|---|
| GEO⁺ (away from Sun) | +42 157 km | $1.4942 \times 10^{11}$ | 500 |
| GEO⁻ (toward Sun) | −42 157 km | $1.4858 \times 10^{11}$ | 500 |
| GPS⁺ (away from Sun) | +26 371 km | $1.4926 \times 10^{11}$ | 500 |
| GPS⁻ (toward Sun) | −26 371 km | $1.4874 \times 10^{11}$ | 500 |
| Earth (reference) | 0 | $1.4900 \times 10^{11}$ | 500 |
| Mars | — | $2.0900 \times 10^{11}$ | 700 |

### 7.2 Complete Temporal Flux Table

| Location | $F(r,t)$ | $\Delta F$/hr (μs/hr) | $\Delta F$/day (μs/day) |
|---|---|---|---|
| Earth (ref.) | $-2.1605 \times 10^{-7}$ | 0.0000 | 0.0000 |
| GEO⁺ | $-2.1599 \times 10^{-7}$ | +0.2200 | +5.2800 |
| GEO⁻ | $-2.1611 \times 10^{-7}$ | −0.2201 | −5.2830 |
| GPS⁺ | $-2.1601 \times 10^{-7}$ | +0.1376 | +3.3032 |
| GPS⁻ | $-2.1609 \times 10^{-7}$ | −0.1377 | −3.3044 |
| Mars ($t_1 = 700$ s) | $-1.5403 \times 10^{-7}$ | +223.29 | +5358.93 |

**Sign convention:** $\Delta F > 0$ → slower clock (time dilation); $\Delta F < 0$ → faster clock. 7.3 GEO Temporal Dilation

| Location | F(r,t) | ΔF/hr (μs/hr) | ΔF/day (μs/day) | Interpretation |
|---|---|---|---|---|
| **Earth (ref.)** | $-2.1605 \times 10^{-7}$ | 0.0000 | 0.0000 | Reference |
| GEO⁺ (away Sun) | $-2.1599 \times 10^{-7}$ | +0.2200 | +5.2800 | Slower clock |

| | | | | |
|---|---|---|---|---|
| GEO⁻ (toward Sun) | $-2.1611 \times 10^{-7}$ | −0.2201 | −5.2830 | Faster clock |
| GPS⁺ (away Sun) | $-2.1601 \times 10^{-7}$ | +0.1376 | +3.3032 | Slower clock |
| GPS⁻ (toward Sun) | $-2.1609 \times 10^{-7}$ | −0.1377 | −3.3044 | Faster clock |
| **Mars ($t_1$=700 s)** | $-1.5403 \times 10^{-7}$ | +223.29 | +5358.93 | Much slower clock |

## 8. Temporal Dilation on Mars

At $t_1 = 700$ s (light propagation time Sun → Mars): $F(r_{\text{Mars}}, t_1) = \frac{A}{4\pi r_{\text{Mars}}^2}[(r_{\text{Mars}} - r_0)(r_{\text{Mars}} - R) + \varepsilon t_1 + c^2 t_1^2]$

$$F(2.09 \times 10^{11}, 700) = -1.5403 \times 10^{-7}$$

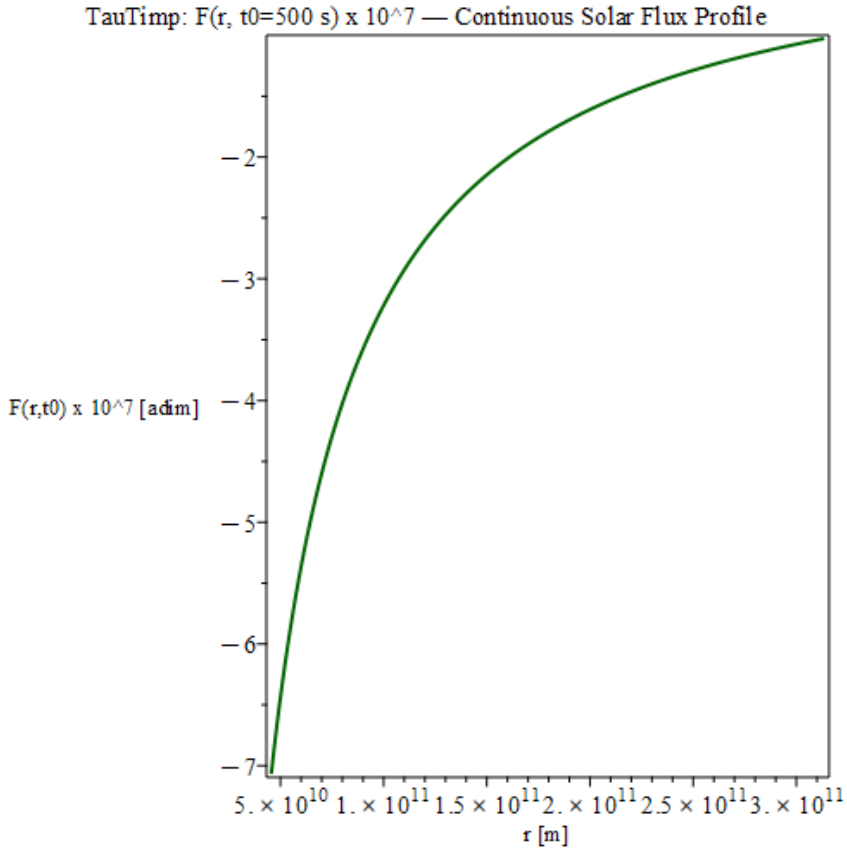
*$\Delta F_{Mars}/hr = +223.29\ \mu s/hr$*

*$\Delta F_{Mars}/day = +5358.93\ \mu s/day \approx 5.36\ ms/day$*

The passage of time on Mars is **significantly slower** than on Earth due to the weaker solar gravitational field at $r_{\text{Mars}}/r_{\oplus} \approx 1.40$, amplified by the cosmological factor $(r - R)$ in the TauTimp ansatz.

## 9. Summary of TauTimp Parameters

| Parameter | Symbol | Value (SI) | Unit |
|---|---|---|---|
| Schwarzschild threshold | $r_0$ | 11 222.3 | m |
| TauTimp cosmological radius | $R$ | $4.2355 \times 10^{22}$ | m |
| Amplitude (GEO calibrated) | $A$ | $9.5510 \times 10^{-18}$ | $m^{-2}$ |
| Linear temporal coefficient | $\varepsilon$ | $1.6610 \times 10^{15}$ | $m^2/s$ |
| Earth time quantum | $\tau_q$ | $1.765 \times 10^{-19}$ | s |
| Speed of light | $c$ | $2.998 \times 10^{8}$ | m/s |
| Sun→Earth propagation time | $t_0$ | 500 | s |
| Sun→Mars propagation time | $t_1$ | 700 | s |
| $A$ (natural units) | $A_{\text{nat}}$ | $3.719 \times 10^{-49}$ | $GeV^2$ |
| $\varepsilon$ (natural units) | $\varepsilon_{\text{nat}}$ | $5.540 \times 10^{21}$ | fm |




The temporal flux is slow-flowing in the Solar system.